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ORIGINAL ARTICLE



Heart rate control is associated with reduced cardiovascular events in Asian patients with coronary artery disease treated with bisoprolol (BISO-CAD): results from a multi-national, real-world experience

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ABSTRACT

Objective: To evaluate the association between decrease in resting heart rate (RHR) and occurrence of composite cardiac clinical outcomes in coronary artery disease (CAD) patients after bisoprolol treatment.

Methods: This phase IV, multi-national, single-arm, open-label, non-randomized, observational trial was conducted between October 2011 and July 2015 across 42 hospitals from China, South Korea and Vietnam.

Results: Analysis of 866 patients (mean age 63.85 ± 10.35 ; mean RHR at baseline 75.71 ± 6.87 bpm in intent-to-treat [ITT]; 75.56 ± 6.73 in efficacy analysis [EA] sets) was performed. Patients with lower mean RHR had fewer composite cardiac events and patients with RHR of 69–74 bpm reported significantly higher outcomes than patients with RHR < 65 bpm ($p = .0449$). A significant association with occurrence of the composite cardiac outcome and hospital admission for unstable angina or revascularization was reported in the EA set (regression estimate: 0.03, 95% CI 0.00–0.07, $p = .0412$) and not in the ITT set for bisoprolol treated CAD patients. Composite cardiac outcomes significantly increased in patients with mean RHR ≥ 70 bpm compared to patients with mean RHR < 70 bpm ($p = .0328$). Adverse events (AEs) were reported in 206 (23.8%) patients, of whom 102 (11.8%) patients had serious adverse event (SAEs). Among the patients with SAEs, 11 (1.3%) patients died. Treatment related adverse events were only 12 (1.4%). No treatment related SAE happened.

Conclusion: The findings showed bisoprolol to be efficacious, in terms of lowering RHR and causing a significant decrease in the occurrence of the composite cardiac outcome, as well as safe in Asian patients with CAD.

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Bisoprolol; coronary artery disease; phase IV

Introduction

Elevated resting heart rate (RHR) is an important risk factor of coronary heart or coronary artery disease (CHD or CAD), which negatively influences the myocardial oxygen balance by increasing consumption and decreasing supply of oxygen^{1,2}. In addition, there is evidence showing the association of elevated RHR with increased risk of hospitalization, cardiovascular (CV) morbidity and mortality in healthy populations³ and patients with CAD^{4–6}. Therefore it is imperative to control the RHR in CVD and maintain it at < 60 beats per min (bpm), which is also recommended for patients with CAD, as suggested by the 2006 European Society of Cardiology guidelines for management of stable angina^{1,7}.

The available evidence corroborates the beneficial effects of beta-blockers (β -blockers) in patients with CAD^{8–10}.

They are considered as the first-line therapy for angina treatment^{8,11,12} and also recommended for non-ST elevation acute coronary syndrome, especially for those with hypertension, tachycardia and long term secondary prevention of angina¹³. Although β -blockers are the first choice drugs for patients with CAD and are accepted by clinicians, multiple studies have reported that RHR was inadequately controlled and remained ≥ 70 bpm in a large percentage of patients with CAD treated with β -blockers^{2,14–19}. Beta-blockers are initiated at low doses, gradually uptitrated to maximum tolerated dose and the treatment is continued long term in patients with CAD²⁰. However, concerns regarding the inappropriate dosing of β -blockers still exist due to physicians' perception of the AEs after treatment with β -blockers²¹. Poor outcomes with under-dosing of β -blockers have

been reported in patients with myocardial infarction (MI)^{22–24}, resulting in increased mortality²⁵. Since RHR is an important marker of cardiac output, appropriate dose selection of β -blockers must be driven by the target RHR for maximizing the benefits in patients with CAD²⁶.

There is much clinical evidence on heart rate (HR) targets and bisoprolol dosage in Western populations^{14–18}, with little evidence in Asian CAD patients^{19,27}. Target RHR in the majority of these studies was ≤ 70 bpm with bisoprolol dose of 2.5 to 5 mg/day. However, the association between varying RHR and prognosis of patients with CAD treated with cardio-selective $\beta 1$ -blockers has not been investigated in CAD patients from the Asia Pacific. Therefore, we conducted this real-world study to assess the association between different RHR and composite cardiac clinical outcomes (i.e. CV death, non-fatal acute myocardial infarction [AMI] and hospital admission for unstable angina [UA] or revascularization) in patients with CAD treated with bisoprolol fumarate (Concor¹) and provide evidence on the improved prognosis of patients with CAD. The findings will also provide the evidence of a positive benefit–risk profile after increasing the dose of bisoprolol to achieve the target RHR.

Methods

Study design

This study was a phase IV, multi-national, single-arm, open-label, non-randomized, observational trial. The trial was conducted in 42 hospitals across China, India, South Korea and Vietnam from 25 October 2011 to 17 July 2015. Due to regulatory restrictions, the trial had to be terminated in India after initiation. It was decided before data base lock that all data from India (50 recruited patients) was not to be analyzed. The patients were followed up for a duration of 18 months in the study and the patients were followed up at 6, 12 and 18 months after baseline.

Study participants

Male and female patients with CAD with stable angina or UA aged ≥ 20 years old, and accompanied with diabetes mellitus (T2DM) or with ≥ 3 risk factors (age ≥ 55 years [males], ≥ 65 years [females], hypertension, dyslipidemia, smoking, family history of CAD and obesity); office RHR ≥ 70 bpm; signing the informed consent before enrolment; with administration of bisoprolol fumarate before enrolment were considered eligible for inclusion in the study. Patients were excluded from the study if they had hypersensitivity or were contraindicated to receive bisoprolol; had AMI in the last 1 month, active myocarditis, severe or uncontrolled hypertension (systolic blood pressure > 180 mmHg or diastolic blood pressure > 110 mmHg), coronary heart failure (CHF, New York Heart Association [NYHA] class III or IV), impaired hepatic or renal function (levels of serum alanine aminotransferase, serum aspartate aminotransferase and serum creatinine twice the normal value upper limit), and any condition considered incompatible by the investigators. Patients who had received complete revascularization or cardiac implant for CAD; had chances of undergoing surgery for valvular disease in the

next 3 years or adding a new drug or undergoing dose adjustment for already prescribed HR altering drugs; pediatrics, and women without effective contraception measures or who were pregnant or lactating were also excluded from the study.

The study protocol was approved by the institutional review boards of all the respective sites in accordance with the International Conference on Harmonization guidelines for Good Clinical Practice (ICH-GCP E6, 1996) and conformed to the Declaration of Helsinki (1964) and its subsequent revisions. All patients received information on the purpose and conduct of this study, and provided written, informed consent.

Efficacy and safety outcomes

Efficacy analysis

The primary study endpoint was occurrence of composite cardiac outcome (inclusive of CV death, non-fatal AMI and hospital admission for UA or revascularization) within 18 months after receiving bisoprolol. Secondary endpoints included assessment of: (i) differences in mean RHR at baseline, 6, 12 and 18 months; (ii) difference in heart function parameters (ejection fraction [EF], fractional shortening [Fs], left ventricular end systolic diameter [LVESD], interventricular septum thickness [IVST], left ventricular posterior wall thickness [LVPWT], E/A ratio and Tei index) at baseline, 6 and 18 months; (iii) carotid intima media thickness (IMT) at baseline, 12 and 18 months; (iv) mortality (all-cause and CV death); and (v) effect of RHR on hospitalization due to acute coronary syndrome (ACS), coronary revascularization and cerebral events during 18 months. Mean HR was defined as the average of all the HRs measured during the 18 months weighted by the actual number of days between the current HR measurement and the previous HR measurement.

Heart rate was measured at baseline, 6 months, 12 months and 18 months during the treatment and was summarized by mean, median, SD, Q1–Q3 and Min–Max. Mean RHR used for primary analyses was the average heart rate which was calculated using all the heart rates measured during the 18 months weighted by the actual number of days between the current heart rate measurement and the previous heart rate measurement. Mean RHR was treated as a categorical variable for the primary analyses.

Sub-group analysis

Further sub-group analysis was performed for both primary and secondary efficacy endpoints. The sub-groups analyzed in the study included: (i) patients aged ≥ 60 years vs. < 60 years, ≥ 70 years vs. < 70 years; (ii) patients with or without history of AMI, HF, T2DM, hypertension and stroke and with history of stable or UA before the trial; (iii) duration of bisoprolol administration starting from baseline (< 6 months, 6 to 12 months and ≥ 12 months).

Exploratory analysis

Exploratory analyses were performed to determine the relationship between the baseline RHR, last RHR, change in RHR,

and duration of bisoprolol usage with the primary and secondary endpoints.

Ad-hoc analysis

The mean dose of bisoprolol at different HR intervals and mean HR of different maximal doses of bisoprolol were evaluated in the ad-hoc analysis. The composite cardiac outcome was analyzed using four different RHR cut-off values (60, 65, 70 and 75 bpm). The occurrence of the composite cardiac clinical outcome was compared among patients with mean RHR <60 and ≥ 60 bpm; <65 and ≥ 65 bpm; <70 and ≥ 70 bpm; and <75 and ≥ 75 bpm.

Safety analysis

Safety outcomes assessed included determination of AEs, discontinuation due to AEs and serious AEs (SAEs). All AEs reported during the study were coded as per the Medical Dictionary for Regulatory Activities (MedDRA) and listed, and their incidence, severity, and relationship to the study drug were tabulated.

Statistical analysis

The sample size calculation was based on the primary endpoint, the composite of CV death, non-fatal AMI, and admission to hospital for unstable angina or revascularization. We assumed the HR distribution of HR to be normal and its relationship with the composite cardiac outcome (y) was described using a Poisson regression model:

$$\text{Log}(y) = \beta_0 + \beta * (\text{HR} - \mu) / \text{SD}$$

where μ and SD are the expected value and standard deviation of the heart rates in the study population.

The SD and β were taken as 10 bpm and 0.15²⁸, which results in an increase of 10 bpm (SD = 1) in HR corresponding to an increase of $\exp(\beta) - 1 = 16\%$ in the expected value of y .

Assuming $\beta_0 = \log(0.45)$, a sample size of 766 evaluable subjects was required to provide 80% power to test the effect of HR at the .05 level. A drop-out rate of 15% was assumed and a total of 902 subjects were required for inclusion in the study. Exclusion of Indian subjects from the analysis reduced the power of the study to 77% and a requirement of 733 evaluable subjects to test the effect of HR on composite cardiac clinical outcomes.

The study participants were analyzed in three different analysis sets: intent-to-treat (ITT), efficacy analysis (EA) and safety analysis (SA) sets. Efficacy analysis was conducted using the ITT and EA sets. Safety analysis was conducted using the SA set. The ITT and SA sets included all the subjects who signed the informed consent and received at least one dose of trial treatment (1.25 mg once daily [QD] to 10 mg QD). Patients without deviation from the inclusion/exclusion criteria and without early termination of the study (with occurrence of primary endpoint) were included in the EA set. All primary and secondary quantitative endpoints were summarized using descriptive statistics. The relationship

between HR and the composite primary endpoint was analyzed using Poisson regression. Continuous secondary endpoints (heart function parameters and IMT, effect of HR) were analyzed via mixed model repeated measures using Statistical Analysis System (SAS) PROC MIXED. Survival analysis was performed using the Cox proportional hazard model. A p -value of <.05 was considered to be statistically significant for the primary analysis. Other analyses were of an exploratory nature. The variables were presented with 95% CIs, wherever applicable. SAS Version 9.1 (SAS Institute Inc., Cary, NC, USA) was used to perform all statistical analysis.

Results

Patient demographics and characteristics

A total of 889 patients with CAD were screened and 23 were excluded after withdrawing informed consent. Therefore, 866 patients were enrolled in this study (ITT set, mean age 63.85 ± 10.35 , mean body mass index [BMI] $25.45 \pm 3.19 \text{ kg/m}^2$). Nearly all the patients ($n = 859$, 99.2%) had history of ≥ 1 disease (17.2% had grade 1, 47.9% had grade 2, 33.4% had grade 3 and 0.7% had grade 4 conditions). Of the study population, 78.6%, 47.5% and 12.4% had hypertension, T2DM and HF, respectively. Mean BP at baseline was $134.3 \pm 15.2 / 78.0 \pm 10.5 \text{ mmHg}$. Mean RHR at baseline was $75.71 \pm 6.87 \text{ bpm}$ and $75.56 \pm 6.73 \text{ bpm}$ in the ITT and EA sets. Other important baseline characteristics are presented in Table 1. The EA set included 688 patients out of the ITT population. A study flow diagram is presented in Figure 1.

Primary endpoint: association between resting heart rate and composite outcome

In the ITT set ($N = 866$), the maximum number of composite outcomes was reported in patients with mean RHR 69–74 bpm (19 in 227 patients), followed by 65–69 bpm (13 in 216 patients), ≥ 74 bpm (12 in 220 patients) and <65 bpm (7 in 203 patients). In patients with higher mean RHR the occurrence of composite cardiac events was greater. In both the ITT and EA sets, the highest incidence of CV events was reported in patients with mean RHR 69–74 bpm ($n = 19$ and 18, respectively) followed by patients with ≥ 74 bpm ($n = 12$ and 11, respectively) (Figure 2(a)). Patients with mean RHR of 69–74 had a significantly higher number of composite cardiac events than patients with mean RHR <65 bpm (ITT: 19 vs. 7 events – regression coefficient estimate (RE) 0.89, SE 0.44, 95% CI: 0.02–1.75, $p = .0449$; EA: 18 vs. 5 events – RE 0.89, SE 0.44, 95% CI: 0.02–1.75, $p = .0180$). Reduction in continuous mean RHR after bisoprolol treatment had no significant effect on the occurrence of the composite cardiac outcome in the ITT (RE 0.02, SE 0.02, 95% CI: 0.00–0.07, $p = .3114$). However, a significant effect was observed in the EA set patients (RE 0.03, SE 0.02, 95% CI: 0.00–0.07, $p = .0412$) undergoing treatment with bisoprolol. In patients aged ≥ 70 years, incidence of CV events increased with mean RHR in both the ITT and EA sets, with incidences being highest in the ≥ 74 and 69–74 bpm groups (Figure 2(b)). Continuous mean RHR and occurrence of composite CV events were

Table 1. Baseline and demographic characteristics.

Demographic Characteristics	Value, N (%)
Total number of patients (ITT), N	866
Gender	
Male	621 (71.7)
Female	245 (28.3)
Age (mean $\pm$ SD)	63.85 $\pm$ 10.35
BMI	
N	828 (95.6)
Mean (kg/m ²)	25.45 $\pm$ 3.19
Age categories	
<60 years	319 (36.8)
$\geq$ 60 years	547 (63.2)
<70 years	607 (70.1)
$\geq$ 70 years	259 (29.9)
Race, Asians	866 (100)
History of 1 disease related to CAD	
Yes	831 (96)
No	28 (3.2)
Unknown	1 (0.1)
Subjects with at least one ongoing medical history	859 (99.2)
Hypertension, n (%)	681 (78.6)
SBP (mmHg)	134.3 $\pm$ 15.2
DBP (mmHg)	78.0 $\pm$ 10.5
Diabetes, n (%)	374 (43.1)
Mean blood glucose (mg/dL)	7.71 $\pm$ 2.45
HF (NYHA grade)	107 (12.4)
Grade 1	24 (2.8)
Grade 2	83 (9.6)
Grade 3	0
Grade 4	0
Angina pectoris, n (%)	87 (10.0)
Unstable angina, n (%)	52 (6.0)
Concomitant drugs used for, n (%)	
Alimentary and metabolic disorders	390 (45.0)
Diabetes	327 (37.8)
CVD (other than bisoprolol)	825 (95.3)
Hematological disorders	810 (93.5)
Concomitant procedures, n (%)	
Percutaneous coronary intervention	3 (0.3)
Coronary artery bypass grafting	1 (0.1)
Per protocol (PP), N	688 (79.4)

Abbreviations. ITT, Intent-to-treat; BMI, body mass index; CAD, coronary artery disease; SBP, systolic blood pressure; DBP, diastolic blood pressure; HF, Heart failure; CVD, cardiovascular disease.

significantly associated in the ≥ 70 years EA set (RE 0.11, SE 0.03, 95% CI 0.04–0.17, $p = .0008$, Table 2).

Secondary endpoints: change in resting heart rate and impact on heart function parameters, intima media thickness, mortality, hospitalization due to acute coronary syndrome and cerebral revascularization

From the mean baseline value of 75.71 ± 6.87 bpm, the mean RHR decreased to 68.98 ± 8.36 bpm at 6 months ($p < .0001$). The mean RHR at 12 months (69.40 ± 8.08 bpm) and 18 months (68.67 ± 9.52 bpm) was also significantly lower compared with the baseline mean RHR ($p < .0001$) showing consistent control in RHR with bisoprolol treatment. Decrease in mean RHR at 6, 12 and 18 months was significant compared with baseline ($p < .0001$ for all comparisons, Figure 3). During the 18 month follow-up, none of the heart function parameters including cardiac systolic and diastolic function parameters as measured by echocardiography changed significantly at 6 and 18 months from baseline in the ITT and EA sets. IMT did not demonstrate significant change at different follow-ups from baseline ($p = .6051$).

Of the 11 (1.3%) deaths occurring during the study, no death event was reported in subjects with mean RHR < 65 bpm. Mean continuous RHR showed a significant effect on all-cause mortality in the ITT (RE 0.07, SE 0.02, HR 1.077, $p = .0024$) and EA (RE 0.07, SE 0.02, HR 1.077, $p = .013$) sets. Continuous average RHR had no significant effect on hospitalization due to ACS in the ITT (RE 0.01, SE 0.02, HR 1.013, $p = .4841$) and EA (RE 0.02, SE 0.02, HR 1.025, $p = .1844$) sets. A similar observation was reported for coronary revascularization in both the ITT (RE 0.04, SE 0.02, HR 1.039, $p = .1399$) and EA (RE 0.05, SE 0.03, HR 1.046, $p = .0742$) sets. Risk of cerebral event occurrence was not impacted by continuous mean RHR (ITT: RE 0.00, SE 0.03, HR 1.002, $p = .9514$; EA: RE 0.01, SE 0.03, HR 1.015, $p = .16308$). There was more than three-fold higher risk for hospital admission for ACS in subjects with HR 69–74 bpm than in subjects with the lowest HRs (< 65 bpm) in EA set subjects.

Sub-group analysis: patients with stable angina and without history of heart failure have significant risk of composite outcome occurrence

Sub-group analysis reported a significant effect of mean RHR on the composite cardiac outcome in patients with a history of AMI, stable angina, hypertension, and stroke in the ITT set. Continuous mean RHR was significantly associated with the occurrence of the composite cardiac outcome in patients with stable angina (RE 0.14, SE 0.05, 95% CI 0.05–0.24, $p = .0042$) and age ≥ 70 years (RE 0.07, SE 0.03, 95% CI 0.01–0.13, $p = .0195$) in the ITT set. In the EA set, RHR and occurrence of the composite cardiac outcome were significantly associated in patients aged ≥ 60 years (RE 0.04, SE 0.02, 95% CI 0.01–0.08, $p = .0217$) and patients without history of heart failure (RE 0.04, SE 0.02, 95% CI 0.00–0.07, $p = .0321$) (Table 2). The ITT set includes all the patients who received even a single dose during the observation. However, inclusion of such patients dilutes the results and may indicate little about the results and outcomes, leading to a difference in the results from the EA set.

Exploratory analysis

Continuous RHR, last RHR, change in RHR and duration of bisoprolol use did not affect the composite cardiac clinical outcome in either analysis set. Bisoprolol use did not impact hospitalization due to ACS, coronary revascularization and occurrence of cerebral events.

Ad-hoc analysis

The overall mean dose of bisoprolol administered was 3.63 ± 1.90 mg and the mean duration of treatment was 523.8 ± 133.6 days. The mean maximal dose of bisoprolol received during the study was $3.848 \text{ mg} \pm 2 \text{ mg}$. Of the study population, 45% and 38.7% of patients were treated with doses of 2.5 and 5 mg and majority of the patients (83.7%) were treated for ≥ 12 months. Among the different HR groups, the minimum dose was administered in patients with mean HR < 65 bpm (3.27 ± 1.57 mg) and the maximum dose

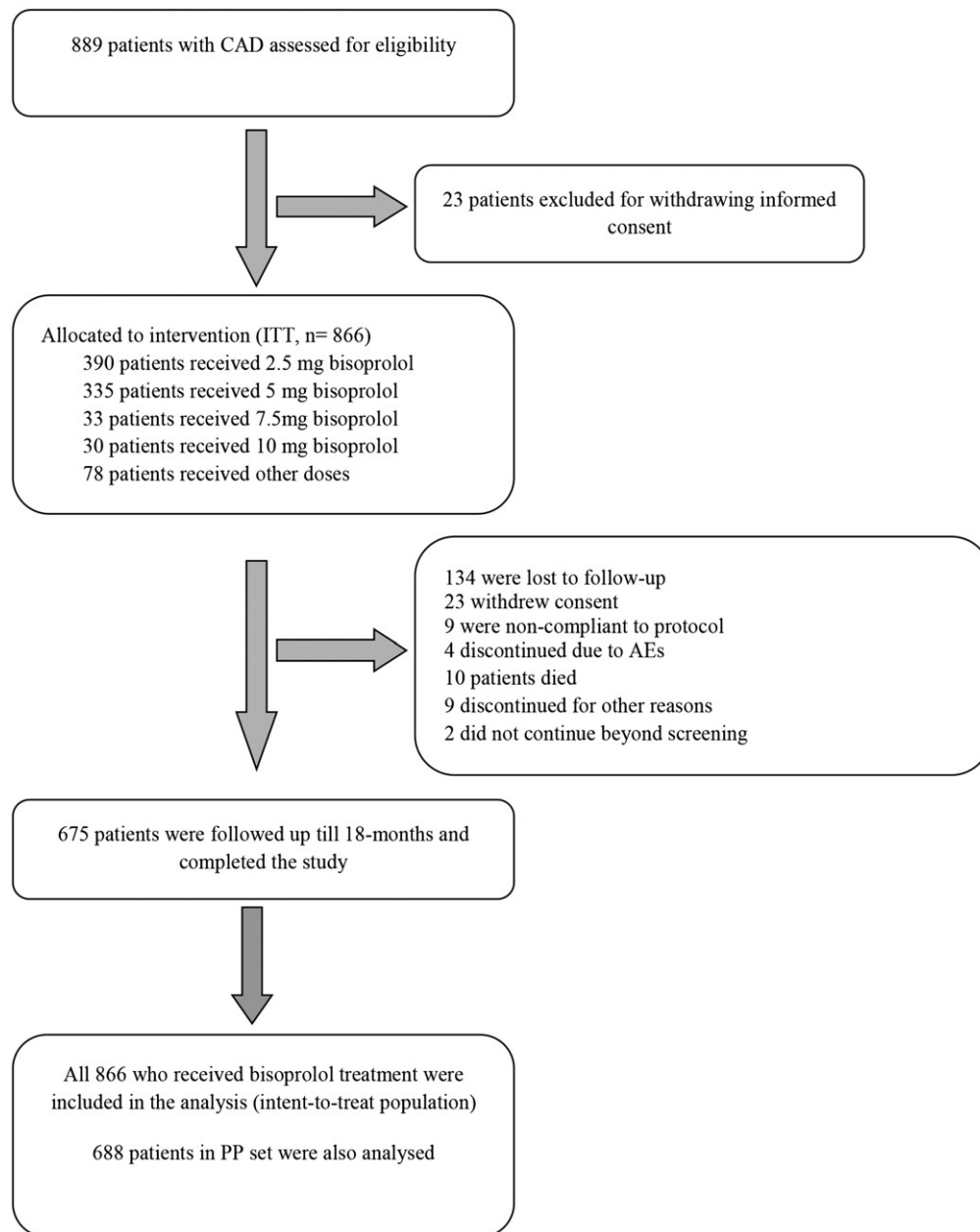


Figure 1. Study flow chart.

was administered in patients with mean HR of ≥ 74 bpm (4.03 ± 2.35 mg). Mean HR at different doses is presented in Supplementary Table 1. As for the effect of mean RHR on the composite cardiac outcome, patients with mean RHR of < 60 and ≥ 60 bpm; < 65 and ≥ 65 bpm and < 75 and ≥ 75 bpm had non-significant differences in composite clinical outcomes in both the ITT and EA sets. Occurrence of CV events was significantly lower in subjects with a mean RHR < 70 bpm than in those with mean RHR ≥ 70 bpm in the EA set ($p = .0328$), but not in the ITT set.

Safety events

The safety assessment reported a total of 206 (23.8%) AEs during the study, of which 12 (1.4%) were considered to be related to bisoprolol treatment. The incidence rate of mild,

moderate and severe AEs was 15.2%, 7.7% and 4.5%, respectively. During the study, the most frequently reported AEs were cardiac disorders (reported in 9.7% of patients), neurological AEs (4.8%), general disorders and administration site conditions (3.8%), metabolic and nutritional nature (2.4%) and gastrointestinal AEs (2.2%), respectively. SAEs were reported in 102 (11.8%) patients, of which 11 (1.3%) resulted in death, 1 (0.1%) was life threatening, 92 (10.6%) required hospitalization or prolongation of hospitalization and 6 (0.7%) included other medically important conditions. No SAE was related to the study drug. Of the 11 (1.3%) deaths, 3 were cardiovascular deaths and the rest were all-cause death (1, metastasis to liver), unknown (1, sudden death), and other (6 including cerebral infarction, chest discomfort, hepatic failure, cardiac arrest, gastrointestinal hemorrhage, and suicide) reasons. All fatal events were unrelated to study drug.

Discussion

As far as we know, there is only a limited number of studies determining the association between RHR control and prognosis in patients with CAD^{4,29} in Western and Asian populations. This real-world study conducted in the Asia Pacific region (China, South Korea, Vietnam and India, however data from India was not included for analysis) provided evidence on HR control and its relationship with chronic stable

CAD after bisoprolol treatment. Further, this study analyzed the impact of mean RHR and continuous mean RHR on composite cardiac outcomes in patients with CAD. In addition, impact on multiple heart function indices and cardiac outcomes were also analyzed in CAD patients belonging to different age groups and having co-morbidities.

Published literature showed high β -blocker usage in a high number of patients with CAD; however, few patients achieved the target RHR^{18,20,21}. We also considered that mean HR (at 6, 12, and 18 months, instead of baseline/end of therapy [EOT] HR) was more appropriate to reflect RHR control and outcomes in patients with CAD. In the CLARIFY registry, patients with higher HR had a greater incidence of CV event occurrence and symptoms of angina²⁹. Similar observations were made in the BEAUTIFUL study where patients with RHR of ≥ 70 bpm reported significantly higher incidences of CV mortality, hospitalization due to HF or MI, and coronary revascularization²⁸. In our study, an increase in the number of composite cardiac events was reported with an increase in RHR. The 69–74 bpm group reported a

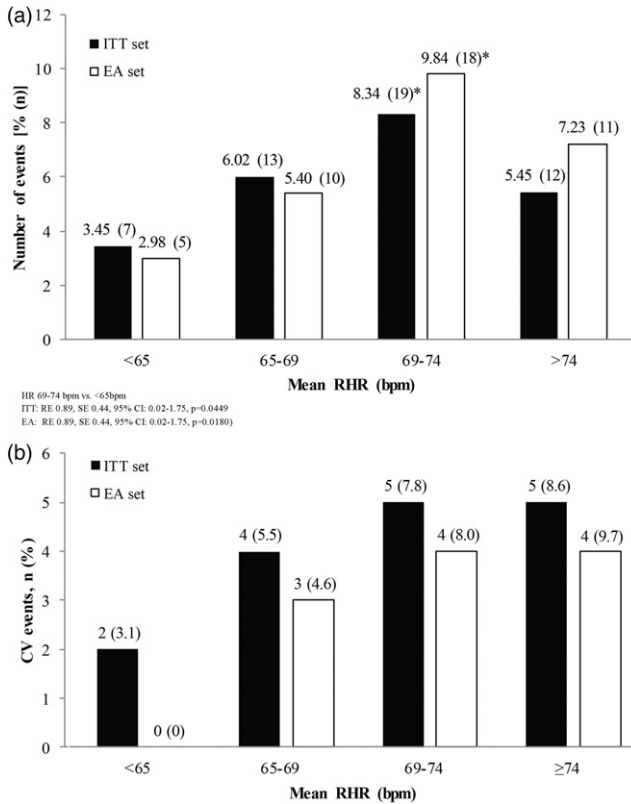


Figure 2. Effect of RHR on composite cardiac endpoint. (a) Incidence of composite cardiac endpoints with different RHRs. (b) Effect of different RHRs in patients aged ≥ 70 years.

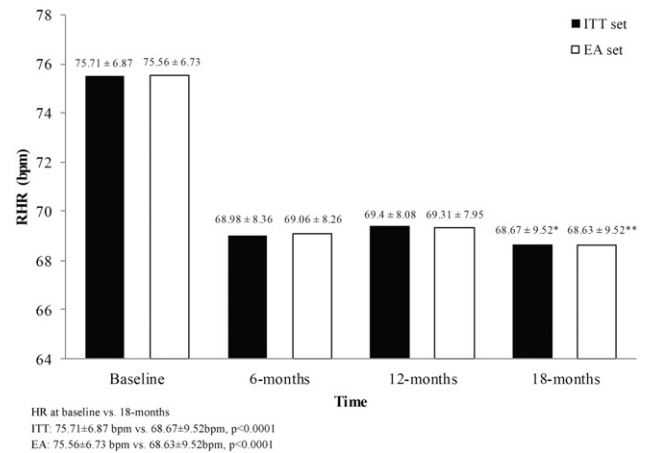


Figure 3. Change in RHR at 6, 12 and 18 month follow-up. *** $p < .0001$ for both ITT and EA – from baseline to 18 months.

Table 2. Effect of continuous mean RHR on composite cardiac clinical outcomes – sub-group analyses.

Analyses*	ITT set		EA set	
	Estimate (SE), [95% CI]	p values	Estimate (SE), [95% CI]	p values
AMI history				
Yes	0.05 (0.05), [-0.04–0.14]	.3202	0.05 (0.04), [-0.04–0.13]	.3080
No	0.01 (0.02), [-0.02–0.05]	.4866	0.05 (0.02), [0.00–0.07]	.0794
Without history of heart failure	0.02 (0.02), [-0.01–0.05]	.2758	0.04 (0.02), [0.00–0.07]	.0321*
With history of stable angina	0.14 (0.05), [0.05–0.24]	.0042	–	–
With history of UA	0.02 (0.03), [-0.04–0.09]	.5022	–	–
With T2DM	0.01 (0.03), [-0.05–0.06]	.8507	0.02 (0.03), [-0.03–0.08]	.3597
Without T2DM	0.02 (0.02), [-0.02–0.07]	.2364	0.04 (0.02), [0.00–0.08]	.0517
With hypertension	0.02 (0.02), [-0.02–0.05]	.3974	0.03 (0.02), [0.00–0.07]	.0631
Without hypertension	0.02 (0.04), [-0.05–0.10]	.5394	0.04 (0.04), [-0.05–0.12]	.4150
Without stroke	0.01 (0.00), [-0.02–0.05]	.4655	0.03 (0.02), [0.00–0.07]	.0807
Bisoprolol use ≥ 12 months	0.02 (0.02), [-0.02–0.06]	.2646	0.03 (0.02), [-0.01–0.07]	.1062
Age				
<60 years	0.00 (0.03), [-0.07–0.06]	.9088	0.01 (0.03), [-0.06–0.07]	.7777
≥ 60 years	0.03 (0.02), [-0.01–0.06]	.1883	0.04 (0.02), [0.01–0.08]	.0217*
<70 years	0.00 (0.02), [-0.04–0.04]	.8784	0.01 (0.02), [-0.03–0.05]	.6701
≥ 70 years	0.07 (0.03), [0.01–0.13]	.0195	0.11 (0.03), [0.04–0.17]	.0008*

*Patient numbers included in analyses on different sub-groups varied.

Composite cardiac clinical outcomes include cardiovascular death, non-fatal acute myocardial infarction and hospital admission for unstable angina or revascularization. Average resting heart rate is calculated using all heart rates measured during the 18 months weighted by the actual number of days between two consecutive measurements. A Poisson regression model is used to assess the effect of resting heart rate on composite cardiac clinical outcomes.

Abbreviations. RHR, resting heart rate; ITT, intent-to-treat; EA, efficacy analysis; SE, standard error; AMI, acute myocardial infarction; UA, unstable angina; T2DM, type 2 diabetes mellitus.

significantly higher number of events in composite CV than reported in the <65 bpm group in both the ITT and EA sets; however, only the EA set population showed a significant association of low mean HR and fewer CV events ($p=.0412$). The findings of our study were in alignment with previous studies with regard to the higher occurrence and correlation of higher RHR with CV events.

The initial dose of β -blockers depended on the RHR at baseline, demonstrating that patients with higher HR are prescribed higher doses of bisoprolol^{30,31}. In our study, patients with higher mean RHR at baseline received a higher dose of the β -blocker bisoprolol fumarate than patients with lower mean RHR. Published literature has revealed that treatment with β -blockers lowered the HR to around <70 bpm^{32–34}. However, many studies have also reported inadequate control in HR (at >70 bpm) in a large percentage of patients after treatment with β -blockers^{14–19}. Inappropriate dosing of β -blockers²¹ and increasing HR are of great concern in elderly patients³⁵. Uptitration of β -blocker dose is limited by achievement of desired HR whereas uptitration is required when the lower doses do not produce sufficient chronotropic response³⁶. Bisoprolol produces a dose dependent change in HR³¹; however, its dose is not optimal in clinical practice due to physicians' perception²¹. In our study, the majority of patients received 2.5 mg/day ($n=390$, 45.0%) and 5 mg/day ($n=335$, 38.7%) doses of bisoprolol indicating the usage of lower doses rather than the higher doses, 7.5 mg/day ($n=33$, 3.8%) and 10 mg/day ($n=30$, 3.5%). Higher mean RHR in patients treated with higher doses (7.5 mg or 10 mg/day) was possibly due to confounding factors that could influence the dose optimization of bisoprolol (Concor as used in this study).

Previous studies of bisoprolol have demonstrated that RHR reduction is an important treatment strategy in improving CV outcomes^{4,27,29–33}. In the MAIN-CHF study II, treatment with beta-blockers (bisoprolol and carvedilol) lowered the occurrence of poor CV outcomes by $>90\%$ post RHR reduction²⁹. Greater RHR lowering with progressive doses of bisoprolol were demonstrated in the CIBIS trial³⁰. In addition, elderly patients (mean age: 72.9 ± 5.4 years) had the lowest mortality rate and fewer treatment emergent AEs when the RHR was controlled within 55 to 64 bpm³¹, showing the positive effect of beta-blocker treatment in the elderly sub-group as well. An analysis of the different doses of bisoprolol (low [up to 3.75 mg/day], medium [5.0 or 7.50 mg/day] and high [10 mg/day]) in the CIBIS II data also confirmed highest survival and lowest occurrence of poor CV outcomes with the medium dose. All doses showed significantly higher survival than placebo³². Our study findings were consistent with the published findings and showed that patients with higher RHR (especially 69–74 bpm) experienced a higher number of composite cardiac outcomes than patients with RHR <65 bpm. In our study, the medium dose of 5.0 to 7.5 mg/day was administered to the highest number of patients followed by the low dose of 2.5 mg/day, which confirmed the findings reported in the CIBIS II study. Furthermore, mean RHR of ≥ 70 bpm was associated significantly with the occurrence of CV events ($p=.0328$). This observation was of particular significance in patients ≥ 70 years old in the ITT ($p=.0195$) and

EA ($p=.0008$) sets, as reported in the sub-group analysis. This suggests that drug treatment with bisoprolol and 70 bpm as the cut-off, especially in patients aged ≥ 70 years, is a feasible option, since CV event occurrence in patients with HR <70 bpm was lower compared to those with HR ≥ 70 bpm. The primary cut-off to <70 bpm might help patients to further achieve the desired target RHR of 55–60 bpm, which is similar to the RHR target recommended by the American Heart Association guidelines and the ESC guideline target of 50–60 bpm^{7,37}. RHR has been shown to be an independent predictor of mortality³⁸ and treatment with β -blockers results in reduced mortality in patients with MI and HF³⁹. Accordingly our study also showed that a reduction in HR with bisoprolol did not cause any mortality in patients with mean RHR <65 bpm. Overall, reduction in mean RHR was significantly associated with lowering all-cause mortality in the ITT ($p=.0024$) and EA sets ($p=.0013$). Bisoprolol improves cardiac function⁴⁰. The majority of patients had normal heart function at inclusion, which was maintained through the study.

There were a few limitations associated with our study. Firstly, a high percentage (15.5%) of subjects was lost to follow-up, which led to a large difference between the ITT and EA sets. Secondly, the follow-up period was limited to 18 months. A longer follow-up, >18 months, would have resulted in evaluation of long-term outcomes with RHR reduction and in CAD patients after bisoprolol treatment. Thirdly, though mean RHR of <70 bpm showed benefit in CAD outcomes, the effect of RHR 50–60 bpm (as recommended by the guidelines) must be evaluated in CAD patients in future studies. Finally, very few patients were treated with higher doses (>5 mg/day) of bisoprolol and the association of higher RHR was not observed with endpoints such as hospitalization due to ACS, coronary revascularization and occurrence of cerebral events; therefore, more real-world studies are required to evaluate the effect of higher doses of bisoprolol on RHR in Asian patients with CAD.

Conclusions

In conclusion, the results of this multicenter, observational study showed that lowering RHR after bisoprolol treatment caused the lowest number of composite cardiac outcomes in Asian patients with CAD and RHR <65 bpm and events increased with higher RHR. In addition, higher continuous mean RHR and age ≥ 70 years significantly increased incidence the of the composite cardiac outcome. Controlling RHR at <70 bpm helped to improve the prognosis in patients with CAD and the dose of bisoprolol needed to be further optimized. Treatment of CAD patients with bisoprolol provided a clinically relevant reduction in the composite cardiac outcome with acceptable safety and tolerability.

Note

1. Concor is a registered trade name of Merck Serono Co. Ltd

Transparency

Declaration of funding

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Declaration of financial/other relationships

Y.C., X.Y., V.N.P., S.H., G.F., X.C., B.Q.T., Y.Y., S.L., X.C., T.M., D.-S.K. and T.K. have disclosed that they have no significant relationships with or financial interests in any commercial companies related to this study or article.

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References

1. Fox K, Borer JS, Camm J. Resting Heart Rate in Cardiovascular Disease. *J Am Coll Cardiol* 2007;50:823-30
2. Balode I, Mintale I, Latkovskis G, et al. Insufficient control of heart rate in stable coronary artery disease patients in Latvia. *Medicina (Kaunas)* 2014;50:295-302
3. Cooney MT, Vartiainen E, Laatikainen T, et al. Elevated resting heart rate is an independent risk factor for cardiovascular disease in healthy men and women. *Am Heart J* 2010;159:612-19.e3
4. Diaz A, Bourassa MG, Guertin MC, et al. Long-term prognostic value of resting heart rate in patients with suspected or proven coronary artery disease. *Eur Heart J* 2005;26:967-74
5. Pieper KS, Gore JM, FitzGerald G, et al. Validity of a risk prediction tool for hospital mortality: the Global Registry of Acute Coronary Events. *Am Heart J* 2009;157:1097-105
6. Kolloch R, Legler UF, Champion A, et al. Impact of resting heart rate on outcomes in hypertensive patients with coronary artery disease: findings from the INternational VERapamilSR/trandolapril Study (INVEST). *Eur Heart J* 2008;29:1327-34
7. Fox K, Alonso Garcia MA, Ardissone D, et al. Guidelines on the management of stable angina pectoris: executive summary. *Eur Heart J* 2006;27:1341-81
8. Anonymous. Effect of metoprolol CR/XL in chronic heart failure: Metoprolol CR/XL Randomized Intervention Trial in Congestive Heart Failure (MERIT-HF). *Lancet* 1999;353:2001-7
9. Ignjatovic V, Pavlovic S, Miloradovic V, et al. Influence of different β -blockers on platelet aggregation in patients with coronary artery disease on dual antiplatelet therapy. *J Cardiovasc Pharmacol Ther* 2016;21:44-52
10. Lin ZP, Dong M, Liu J. Bisoprolol improved endothelial function and myocardium survival of hypertension with stable angina: a randomized double-blinded trial. *Eur Rev Med Pharmacol* 2013;17:784-801
11. Fraker T, Fihn SD. 2007 chronic angina focused update of the ACC/AHA 2002 guidelines for the management of patients with chronic stable angina. *Circulation* 2007;116:2762-72
12. Chinese Society of Cardiology. Guidelines for diagnosis and management of chronic stable angina. *Chin J Cardiol* 2007;35:195-206
13. Chinese Society of Cardiology. Guidelines for diagnosis and management of unstable angina and non-ST segment myocardial infarction. *Chin J Cardiol* 2007;35:295-304
14. Alcocer Gamba MA, Martínez Sánchez C, Verdejo Paris J, et al. Heart rate and use of β -blockers in Mexican stable outpatients with coronary artery disease. *Arch Cardiol Mex* 2015;85:270-7
15. Tendera M, Fox K, Ferrari R, et al. Inadequate heart rate control despite widespread use of beta-blockers in outpatients with stable CAD: findings from the international prospective CLARIFY registry. *Int J Cardiol* 2014;176:119-24
16. Stepińska J, Marona M, Greenlaw N, et al. Heart rate and the use of beta-blockers in stable outpatients with coronary artery disease: Polish baseline results of the CLARIFY registry. *Kardiologia Pol* 2014;72:1156-64
17. Steg G, Ferrari R, Ford I, et al. Heart rate and use of beta-blockers in stable outpatients with coronary artery disease. *PLoS One* 2012;7:e36284
18. Sbarouni E, Voudris V, Georgiadou P, et al. Heart rate and B-blockade in stable coronary artery disease in Greece. *Hellenic J Cardiol* 2015;56:11-27
19. Sun YH, Yu JM, Hu DY. Status of β -blocker use and heart rate control in Chinese patients with stable coronary artery disease. *Zhonghua Xin Xue Guan Bing Za Zhi* 2016;44:19-26
20. Boudonas GE. B-Blockers in coronary artery disease management. *Hippokratia* 2010;14:231-5
21. Squire IB, Barnett DB. The rational use of beta-adrenoceptor blockers in the treatment of heart failure. The changing face of an old therapy. *Br J Clin Pharmacol* 2000;49:19
22. Goldberger JJ, Bonow RO, Cuffe M, et al. Beta-blocker use following myocardial infarction: low prevalence of evidence-based dosing. *Am Heart J* 2010;160:435-42.e1
23. Goldberger JJ, Bonow RO, Cuffe M, et al. Effect of beta-blocker dose on survival after acute myocardial infarction. *J Am Coll Cardiol* 2015;29:66:1431-41
24. Krumholz HM, Radford MJ, Wang Y, et al. National use and effectiveness of beta-blockers for the treatment of elderly patients after acute myocardial infarction: National Cooperative Cardiovascular Project. *JAMA* 1998;280:623-9
25. Ruta J, Ptasiński P, Maciejewski M, et al. Effect of low doses of metoprolol, bisoprolol and carvedilol on mortality in patients with left ventricular dysfunction after acute myocardial infarction. *Wiad Lek* 2006;59:649-53
26. Porapakham P, Porapakham P, Krum H. Is target dose of beta-blocker more important than achieved heart rate or heart rate change in patients with systolic chronic heart failure? *Cardiovasc Ther* 2010;28:93-100
27. Vitale C, Iellamo F, Volterrani M, et al. Heart rate control in an unselected consecutive population of outpatients with stable coronary artery disease: analysis of the CARDIf Study Cohort. *Angiology* 2010;61:763-7
28. Fox K, Ford I, Steg PG, et al. Heart rate as a prognostic risk factor in patients with coronary artery disease and left-ventricular systolic dysfunction (BEAUTIFUL): a subgroup analysis of a randomized controlled trial. *Lancet* 2008;372:817-21
29. Steg PG, Ferrari R, Ford I, et al. Heart rate and use of beta-blockers in stable outpatients with coronary artery disease. *PLoS One* 2012;7:e36284
30. Dungen HK, Bright LM, Inkrot S, et al. Heart rate following short-term beta-blocker titration predicts all-cause mortality in elderly chronic heart failure patients: insights from the CIBIS-ELD trial. *Eur J Heart Failure* 2014;16:907-14
31. Simon T, Krause MM, Brentano CF, et al. Bisoprolol dose-response relationship in patients with congestive heart failure: a subgroup analysis in the cardiac insufficiency bisoprolol study (CIBIS II). *Eur Heart J* 2003;24:552-9
32. Yamashita T, Inoue H. Heart rate-reducing effects of bisoprolol in Japanese patients with chronic atrial fibrillation: results of the MAIN-AF study. *J Cardiol* 2013;62:50-7

33. CIBIS Investigators and Committees. Randomized trial of beta-blockade in heart failure. The Cardiac Insufficiency Bisoprolol Study (CIBIS). *Circulation* 1994;90:1765-73
34. Lechat P, Hulot JS, Escolano S, et al. Heart rate and cardiac rhythm relationships with bisoprolol benefit in chronic heart failure in CIBIS II trial. *Circulation* 2001;103:1428-33
35. Stern S, Behar S, Gootlieb S. Aging and diseases of the heart. *Circulation* 2003;108:e99-101
36. Gelbrich G, Edelmann F, Inkrot S, et al. Is target dose the treatment target? Up-titrating beta-blockers for heart failure in the elderly. *Int J Cardiol* 2012;155:160-6
37. McInnes G, Padmnabha S, Menown I. Heart rate as a risk factor in cardiovascular disease. *Prescriber* 2010;21:45-9
38. Woodward M, Webster R, Murakami Y, et al. The association between resting heart rate, cardiovascular disease and mortality: evidence from 112,680 men and women in 12 cohorts. *Eur J Prev Cardiol* 2014;21:719-26
39. Bunch TJ, Muhlestein JB, Bair TL, et al. Effect of beta-blocker therapy on mortality rates and future myocardial infarction rates in patients with coronary artery disease but no history of myocardial infarction or congestive heart failure. *Am J Cardiol* 2005;95:827-31
40. Issa V, Guimarães GV, Coimbra Rezende MV, et al. Effects of bisoprolol on cardiac function and exercise in patients with heart failure. *Arq Bras Cardiol* 2007;88:307-11

Notice of correction

Please note that Figure 1 has been edited since the paper was first published (1 September 2017).